

Land Art as Architecture: A Sustainable Approach to Place-Making and Environmental Design

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Abstract

Land art, traditionally viewed as an artistic intervention in the landscape, has evolved into a powerful medium for creating architectural space that is both environmentally sustainable and deeply connected to its surroundings. This paper explores how land art can transcend its artistic origins to create architectural spaces that harmoniously integrate with the landscape, offering a sustainable alternative to conventional built environments. By responding to site-specific conditions, and embracing landscape design principles, land art fosters an immersive spatial experience that enhances a sense of place. Through a critical analysis of key land art projects and their spatial characteristics, the study examines how these interventions define place character, frame vistas, and facilitate human interaction with the landscape. Case studies of works by artists such as A. Goldsworthy, R. Smithson, A. Burri, etc and contemporary architects illustrate how land art functions as an architectural form that embodies sustainability, minimal environmental impact, and cultural significance. The findings suggest that land art can serve as a model for future architectural practices, promoting a symbiotic relationship between the built and natural environments while reinforcing identity and belonging within a place. This research ultimately positions land art as an alternative design strategy that redefines architecture offering innovative pathways for sustainable place-making.

Keywords

land art, landscape architecture, sustainable design, place making

Introduction

Land art is a genre of visual art, emerged during the 1960s, that used the natural landscape as a canvas to put outdoor installations. It challenged conventional art by moving artworks outside galleries, in the natural settings (Beardsley, 1998). This genre of art explores the relationship between humans and the land, and often addresses ecological themes such as environmentalism, sustainability, and the human impact on the Earth. One of the defining aspects of land art is how it responds directly to the unique features of a site—its terrain, geology, vegetation, etc. This site-specific focus is similar to the goals of landscape architecture, which objects respond to specific conditions of place, and aim to create designs that are meaningful and environmentally sensitive. As the need for sustainability is becoming a crucial issue, considering land art as architecture is more significant than ever. Today's architectural practice places a growing emphasis on sustainable approaches that reduce environmental impact and enrich the experience of space. Similarly, land artists consider the cultural and ecological context of each place, prompting us to think more deeply about how we interact with the natural world (Wang, 2023; Selanon & Cui, 2020).

In addition to environmental issues, land art addresses also cultural narratives such as the concept of place, identity, and anthropocentrism (Rigaud, 2012). As artist Marlene Creates expresses, *"The land is important to me, but even more important is the idea that it becomes a 'place' because someone has been there"* (in Lippard, 1997: 32). This statement captures the role of human presence in giving meaning to landscapes. Through minimal gestures land art invites to deeply engage with nature. This idea also aligns with architecture, which has the power to transform an empty space into a place filled with meaning, memory, and a sense of belonging. The philosophical ideas at the bases of land art are valid also for landscape architecture, emphasizing themes like reflection, memory, and the lived experience of place (Dang, 2021). This approach has blurred the boundaries between art and architecture. Once a radical artistic intervention, land art now plays a role in placemaking—contributing to the design of environments that are ecologically responsible and culturally resonant. As sustainability becomes a defining concern in architecture, land art offers useful strategies for integrating ecological awareness and spatial meaning into the built environment.

Methodology

This research adopts a qualitative methodological framework that integrates theoretical inquiry, case study analysis, and visual examination to explore the architectural potential of land art as a strategy for sustainable place-making. The study aims to synthesize conceptual insights from the fields of art history, landscape architecture, and environmental design with the practical and spatial strategies evident in selected works of land art and architecture. The methodological approach unfolds in three interrelated phases:

Theoretical Framework Construction

The research begins with an extensive literature review that critically engages with the conceptual foundations of land art, particularly its roots in conceptual art, phenomenology, and ecological thought. Key theoretical perspectives, including those of Lucy Lippard, Robert Smithson, Miwon Kwon, Christian Norberg-Schulz, and contemporary environmental aesthetics

scholars, provide the grounding for understanding land art not only as an artistic movement but as an architectural practice with implications for sustainability and place-making. This phase establishes the ontological and epistemological lens through which the subsequent analyses are conducted.

Case Study Selection and Comparative Analysis

Building on the theoretical foundation, the research undertakes a comparative analysis of selected case studies drawn from both land art and architectural practice. These include seminal land art interventions such as Robert Smithson's *Spiral Jetty* (1970), Michael Heizer's *Double Negative* (1969–70), Andy Goldsworthy's *Walking Wall* (2019), among others, as well as architectural projects by Herzog & de Meuron, SANAA, Maya Lin and others.

Visual and Spatial Analysis

Each case study is further examined through visual documentation and experiential descriptions. This analysis seeks to understand how spatial and material strategies are employed to engage viewers, frame environmental phenomena, and produce affective experiences of place.

The methodology emphasizes a dialogical reading of land art and architecture, treating them not as separate disciplines but as overlapping practices that co-produce meaning in landscape. By synthesizing findings across cases, the study identifies recurring patterns, aesthetic logics, and design principles that inform a broader understanding of how land art contributes to sustainable and culturally resonant architectural practice. The research concludes with a critical reflection on the transformative potential of land art in contemporary design, emphasizing its value in fostering ecological awareness, reinforcing site identity, and enriching the spatial and emotional experience of place.

Literature review

Land Art as Conceptual art

Land Art emerged in the late 1960s as a radical shift from traditional art forms, within the broader movement of Conceptual Art, which prioritizes ideas over objects. In *The Dematerialization of Art* (1968), Lucy Lippard and John Chandler described this shift toward artworks that focus on process, context, and meaning. Within this framework, Land Art investigates the relationship between humans and the natural environment, and in response, create actions, changes, often through site-specific interventions that transform outdoor landscapes. But Land Art is not merely conceptual in the sense of privileging the idea. It deals with temporality and entropy and the instability of form as highlighted in Robert Smithson's concept of entropy, which rejects classical notions of order and control. Instead of trying to control nature, Land Art works with natural processes like decay and change, showing that both the artwork and the site are unstable and always shifting and highlighting their blurring boundaries of the environment. Miwon Kwon, in *One Place After Another* (2002), redefines the "site" as a network of social, political, and cultural meanings, emphasizing the role of the context in shaping the artwork's significance. In this sense the environment becomes an active part of the creation and interpretation of the artwork. Land Art also resonates with phenomenological philosophy. Maurice Merleau-Ponty's (1962) ideas on embodiment highlight how these works require physical and temporal engagement, making the viewer an active participant. Land Art thus amplifies the experiential

and relational dimensions of Conceptual Art, extending them into the terrain of landscape and ecology. Scholars like Jeffrey Kastner and Brian Wallis (Land and Environmental Art, 1998) underscore how Land Art brings ecological consciousness to the foreground, shifting from human-centered to ecologically entangled modes of thinking. In essence, Land Art is not simply about monumental gestures in the landscape. It is an intellectual and philosophical practice that explores impermanence, place, and ecological interrelation.

Land Art and Environmental Sustainability

Land Art, as a form of artistic and spatial practice in the landscape, offer a critical response to environmental degradation of industrial society. Rooted in deep ecology (Naess, 1973), it emphasizes the need for ecological awareness, sustainability, and low-impact design. According to Arnold Berleant (1992) and Allen Carlson (2000), this engagement with nature enhances how we perceive and ethically interact with the environment. Land Art encourages a multisensory understanding and respect for natural processes. While early Land Art involved large-scale earthworks that altered landscapes, contemporary practice emphasizes the need for more ecological responsibility and site sensitivity. As Suzi Gablik argues in *The Reenchantment of Art* (1991), artists are increasingly called to foster ecological and cultural healing rather than formal innovation. This shift reflects Land Art's evolving commitment to sustainability, moving from a confrontational approach to one of collaboration with nature. Overall, Land Art provides a valuable perspective for linking art and ecology. By working with natural processes and minimizing environmental impact, it plays a key role in promoting sustainability and is an important part of modern discussions on environmental design.

Land Art and Place-Making

Place-making is an important concept in architecture. It focuses on creating spaces that have meaning and identity. Land Art supports this idea by turning ordinary spaces into place through experiential, site-specific interventions. Christian Norberg-Schulz (1980) explains that a place is not just a physical space, but something we feel and experience, shaped by nature and human-made elements. Land Art helps create such places by working with the land's shape, direction, materials, and size to encourage meaningful and sensory experience. As Kastner and Wallis (1998) note, these interventions alter perception and deepen the human-landscape relationship. Lucy Lippard (1997) further argues that Land Art invites bodily interaction, allowing land to be navigated and reinterpreted in poetic ways. Like architecture, it combines aesthetic, spatial, and cultural dimensions, contributing meaningfully to the creation of place.

Case Studies: Analysis and Discussion

Building upon the theoretical frameworks established in the literature, this section examines a series of key case studies in both land art and architecture to reveal a shared vocabulary of environmental sensitivity, site-specificity, and experiential place-making. Through an analysis of works by artists such as Robert Smithson, Michael Heizer, Agnes Denes, Richard Long, Andy Goldsworthy, Nancy Holt, alongside architects including Maya Lin, Herzog & de Meuron, Peter Eisenman, Ryue Nishizawa, SANAA, Louis Kahn, and others, we examine how conceptual art practices have influenced architectural thinking, particularly regarding ecological engagement,

temporal processes, and phenomenological experience. Robert Smithson's *Spiral Jetty* (1970) (Fig. 1), constructed on the shores of the Great Salt Lake in Utah, is a foundational work in land art that reconfigures the traditional understanding of form and site. His massive spiral emerges and disappears with the lake's fluctuating water levels, emphasizing entropy, geological time, and the mutability of human intervention in the landscape (Iorga, 2013; Rigaud, 2012). Smithson's engagement with natural processes reflects a non-anthropocentric view of design, where form is not fixed but in dialogue with environmental forces. His approach has a similar concept to the Herzog & de Meuron's *Dominus Winery* (2000) in Napa Valley) (Fig. 2), which features gabion walls filled with local stone that regulate light and temperature. The structure performs as a porous, breathing organism, integrating the architecture with its geological context. Both works challenge static formalism, opting instead for material and experiential strategies that reflect nature's temporality and the embeddedness of place.



Figure 1. Robert Smithson's *Spiral Jetty* (1970), *Figure 2.* Herzog & de Meuron's *Dominus Winery* (2000)

Michael Heizer's *Dissipate*, (1968–70) (Fig. 3), carved into the edge of the Mormon Mesa in Nevada, interrogates the scale of human intervention in the landscape and the ideological implications of land manipulation. By using subtraction as tool for engaging with memory and place, Heizer removed tons of rock to create two massive voids, confronting viewers with absence as presence, and creation of meaning. The emptiness becomes the medium through which viewers confront scale, memory, and the ideological implications of altering land. A conceptual parallel emerges in Alberto Burri's *Il Cretto di Gibellina* (1984–89) (Fig. 4,5), which overlays the earthquake-destroyed Sicilian town of Gibellina with a concrete shroud, preserving the street grid as a fossilized scar. Both interventions transform trauma into spatial narrative, using voids and scars to inscribe memory and history into the land (Bae-Dimitriadis, 2021). Their monumental scale reclaims loss and silence as architectural expressions, foregrounding the landscape itself as a medium of remembrance.

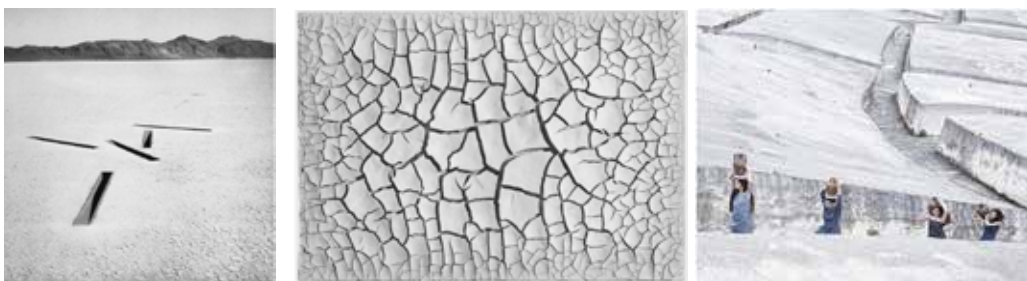


Figure 3. *Dissipate*, 1968–70, by Michael Heizer ,
Figure 4. Alberto Burri, *Cretto - Installazione*,
Figure 5. *Il Cretto di Gibellina* (1984–89) by Alberto Burri



*Figure 6, Wheatfield – A Confrontation (1982) by Agnes Denes,
Figure 7 Eden Project (2001) by Grimshaw Architects*

Artists like Agnes Denes expand the conceptual boundaries of land art by considering the landscape as a cultural and ecological canvas and including concepts of sustainability, public engagement, and visual activism (Bae-Dimitriadis, 2021; Dang, 2021). In her landmark piece *Wheatfield – A Confrontation* (1982), Denes plants a field of wheat on a landfill in downtown Manhattan, juxtaposing nature and culture, sustainability and capitalism, provoking questions about land use, food production, and urban priorities (Dang, 2021). This performative landscape challenges viewers to reconsider the use of public space and the values embedded in the urban fabric. A comparable architectural gesture is found in the *Eden Project* (2001) by Grimshaw Architects, which rehabilitates a former clay pit into a regenerative site of ecological education. The biomes house diverse ecosystems, visually and symbolically contrasting organic life with the constructed geodesic structures as industrial remnants. In both works, the landscape becomes a transformative agent—capable of healing, educating, and critiquing dominant narratives of progress and productivity.



*Figure 8. A Line Made by Walking (1967) by Richard Long,
Figure 9. Walking Wall (2019) Museum of Art in Kansas City by Andy Goldsworthy's,
Figure 10. Vietnam Veterans Memorial (1982) by Maya Lin*

Minimal gestures by Richard Long likewise redefine the landscape through movement and materiality. Long's *A Line Made by Walking* (1967) turns a transient act into a spatial inscription

and invited viewers to contemplate the passage of time and the act of walking itself (Fig. 8). Similarly, Goldsworthy's *Walking Wall* (2019), a dry-stone wall that physically moved across the grounds of the Nelson-Atkins Museum in Kansas City, redefines the role of built form (Fig. 9). As it "walked," the wall traversed roads, lawns, and architectural spaces, becoming a living installation that shifted perceptions of permanence and transition. Importantly, Goldsworthy's work also challenges anthropocentric authorship by emphasizing natural processes: gravity, weather, and decay are co-authors in the work's evolution. These ephemeral acts resonate with architectural practices that prioritize temporality, movement, and the lived experience of space. Maya Lin's *Vietnam Veterans Memorial* (1982) (Fig. 10), ssimilarly transforms minimal form into profound emotional resonance. Sunken into the earth, the black granite walls bear the names of over 58,000 soldiers, creating a tactile, reflective experience. As visitors descend the path and see their own reflections among the names, memory becomes spatialized and embodied. Lin's memorial, like Long's line and Goldsworthy's wall, through a simple gesture, invoke collective memory, identity, and healing. Each of these works underscores the power of landscape—natural and constructed—as a site of intimate encounter and cultural reflection.



Figure 11. Sun Tunnels, 1976 by Nancy Holt, Figure 12. Salk Institute, 1975 by Louis Kahn
Figure 13. The Teshima Art Museum, 2010 by Ryue Nishizawa and Rei Naito

Nancy Holt's *Sun Tunnels* (1973–76), situated in the remote Great Basin Desert of Utah (Fig.11), is a work of land art that responds directly to its context. Four massive concrete cylinders are oriented to frame the sun at solstices and pierce the desert sky with starlike constellations (Tiberghien, 1995). Their openings frame the sky and the desert horizon, while small holes drilled into the surface of the tunnels map out constellations, projecting starlike patterns of light onto the interior surfaces. Holt's intervention is minimal, yet it transforms the emptiness of the landscape in a place. The viewer's position within the tunnels becomes a site of cosmological awareness, merging personal experience with universal cycles.

A parallel architectural gesture is seen in Louis Kahn's *Salk Institute* (1959–65) in La Jolla, California (Fig.12). Framing the Pacific Ocean with two symmetrical laboratory wings, Kahn creates a central plaza where a narrow water channel draws the eye toward the horizon (Brownlee & De Long, 1991). The interplay of light, shadow, and ocean breezes becomes integral to the experience of the place. Both Holt and Kahn celebrate light, sky, and orientation, not as abstract concepts but as guiding forces that give meaning to place.

Ryue Nishizawa's *Teshima Art Museum* (2010) (Fig.13), designed with artist Rei Naito, furthers this integration of natural elements and spatial perception. Located on Japan's Teshima Island,

the museum consists of a single, curved concrete shell that merges with the landscape. Its minimalist form is punctured by circular openings that admit light, air, and rainwater, blurring the boundaries between inside and outside. As water droplets move into the smooth floor, they become ephemeral artworks—reaffirming the museum as a living environment. This project exemplifies a sensorial architecture that prioritizes human perception, seasonal variation, and ecological awareness. In contrast to static form, the Teshima Museum is always becoming—shaped by climate, time, and nature (Tanaka, 2010). In this way, Nishizawa’s Museum aligns with work of Holt and Kahn, both of which also use their designs to frame and enhance a direct, sensory connection with the environment—whether through celestial alignments or by framing panoramic views of the horizon (Frampton, 1995; Tiberghien, 1995).



Figure 14. The Floating Piers 2014-2016, by Christo and Jeanne-Claude

Figure 15. Serpentine Gallery Pavilion 2009, London, by SANAA

This ephemeral approach to material and perception continues in Christo and Jeanne-Claude’s The Floating Piers (2016) (Fig.14), which transforms Lake Iseo into a collective, walkable surface. Visitors traverse water in a surreal, sensory experience that collapses the boundary between built and natural. SANAA’s Serpentine Pavilion (Fig.15), similarly floats above its site with a lightweight reflective canopy that interacts with weather, light, and human movement (Koolhaas & Obrist, 2010). Both projects offer reversible, low-impact interventions that frame the landscape rather than dominate it, fostering ecological awareness through subtle transformation. In both cases, architecture and land art converge not to build permanence, but to temporarily transform landscape into meaningful, sensorial, and poetic place.



Figure 16. Labirinto di Arianna, Sicily by Italo Lanfredini

Figure 17. Memorial to the Murdered Jews of Europe by Peter Eisenman and Buro Happold.

Figure 18. Reiulf Ramstad Arkitekter, Trollstigen Visitor Centre (2009)

Italo Lanfredini's *Labirinto di Arianna* in Sicily (Fig.16) further exemplifies land art's introspective capacity. Rooted in myth, the earthen labyrinth invites tactile, meditative passage, integrating ritual, memory, and self-discovery within the land. Rather than framing the site, it integrates into it, cultivating a sense of ritual, memory, and self-discovery. Its use of natural materials and minimal intervention reflects a deep ecological consciousness in this intervention. This immersive engagement with site and symbolism was used by Peter Eisenman in his *Memorial to the Murdered Jews of Europe* (2005) (Fig.17), where a grid rectangular concrete blocks produces bodily disorientation and spatial ambiguity. Despite its urban setting and architectural rigor, the memorial functions phenomenologically, using absence and repetition to evoke trauma and reflection (Young, 2016). The memorial's restraint—formally minimal and materially austere—further enhances its psychological resonance and spatial depth, aligning with land art's focus on experience over representation.

The *Trollstigen Visitor Centre* by Reiulf Ramstad Arkitekter (Fig.17) extends these principles within a contemporary architectural and infrastructural framework. Set within the dramatic topography of Norway, the centre is embedded in the landscape both conceptually and materially. Corten steel and concrete structures echo the site's geology and weathering patterns, choreographing views and movement through a spatial narrative that highlights scale and temporality (Mollerup, 2013). Its ecological approach lies in its integration and durability, allowing the architecture to age with the site rather than dominate it. Together, these projects—demonstrate a shared approach to landscape as a medium of meaning, not backdrop. Through phenomenological engagement, material sensitivity, and ecological respect, they embody a convergence of land art and architecture, where space is not only seen but profoundly felt.

Conclusion

This study demonstrates how Land Art, as a conceptual and ecological practice, intersects meaningfully with contemporary architectural thought. Through a shared vocabulary of site-specificity, material engagement, and temporal awareness, both land artists and architects challenge conventional notions of form, permanence, and space. The examined case studies illustrate how landscape becomes a medium for reflection, memory, and ecological consciousness. Artists such as Heizer, Denes, and Goldsworthy, alongside architects like Herzog & de Meuron, Grimshaw, and SANAA, foreground natural processes, sensory experience, and contextual responsiveness in their work. These interventions not only engage with place as a physical site but as a cultural, political, and temporal construct.

By transforming space into place, and matter into meaning, Land Art offers critical insights into sustainable design, environmental ethics, and phenomenological engagement. As such, it provides not just aesthetic inspiration, but a conceptual and ethical model for architecture that aspires to be regenerative, place-based, and deeply attuned to the rhythms of nature.

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